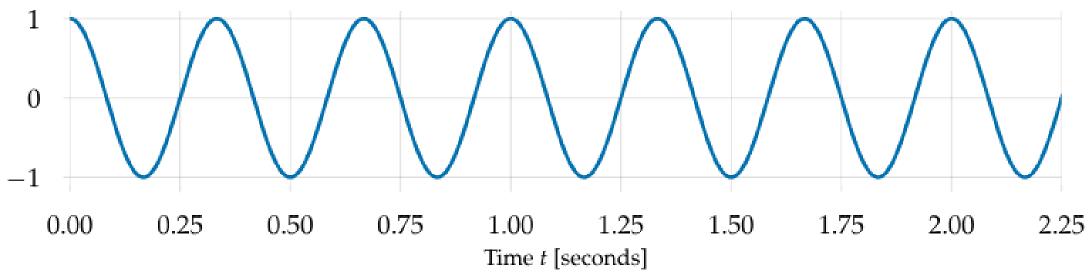
lecture notes 15

Really great reference here: https://brianmcfee.net/dstbook-site/content/intro.html

Fourier transform refresher

Let's say we have a signal $x(t)$:



This signal is periodic if $x(t) = x(t + t_0)$ for some t_0.

Periodic or close to periodic things happen very often in nature! Mostly two categories:

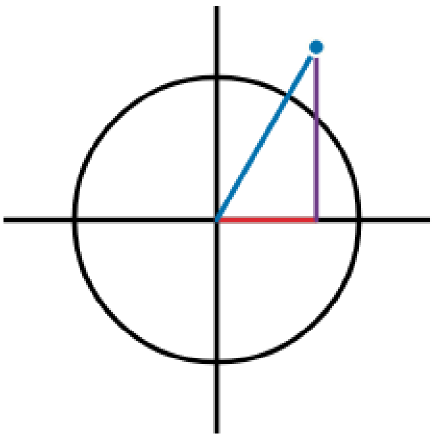
- rotations (orbits, clocks, etc)
- waves (vibrations / sound, ocean, gravitational, etc)

We can write any sinusoid like

$$x(t) = A\cos(\omega t + \phi)$$

Another way to write it is with complex numbers! Remember that $e^{i\theta} = \cos\theta + i\sin\theta$. So we could just as well write the same signal as

$$\begin{aligned} x(t) &= \Re(Ae^{i\omega t + i\phi}) \\ &= \frac{A}{2}(e^{i\omega t + i\phi} + e^{-i\omega t - i\phi}) \end{aligned}$$



A weird (improper!) integral

You guys have probably seen this before, and definitely used it before, but maybe not quite like this. What is

$$
\begin{aligned}
\int_{-\infty}^{\infty} e^{i\omega t} d\omega &= ? \\
&= \lim_{\Omega\to\infty} \left[\frac{e^{i\Omega t}}{i\Omega} + \frac{e^{-i\Omega t}}{i\Omega} \right] \\
&\equiv \delta(t)
\end{aligned}
$$

Normal calculus rules would say that this is **undefined**! This function is oscillating at infinity, so it has no limit. Let's instead consider these oscillations to "cancel out" and give us an integral of 0.

Then the only way this integral could be nonzero is when $t = 0$, then the integrand is just 1, and the integral $\to \infty$. This is not a function, but let's call it the **Dirac delta function**. (The mathematically correct way to do this is not in terms of functions but distributions.)

This "function" has the nice property

$$
\int_{-\infty}^{\infty} \delta(t - t') f(t) dt = f(t')
$$

The Fourier transform
Consider the integral

$$
\int_{-\infty}^{\infty} f(t) e^{-i\omega t} \, \mathrm{d}t
$$

We're basically integrating $f(t)$ against $\delta(\omega)$! One way to see this: take sinusoidal $f(t) = e^{i\omega_s t + i\phi}$

$$
\begin{aligned}
\int_{-\infty}^{\infty} e^{i\omega_s t + i\phi} e^{-i\omega t} \, \mathrm{d}t &= \int_{-\infty}^{\infty} e^{i\omega_s t + i\phi - i\omega t} \, \mathrm{d}t \\
&= e^{i\phi} \int_{-\infty}^{\infty} e^{i(\omega_s - \omega)t} \, \mathrm{d}t \\
&= e^{i\phi} \delta(\omega_s - \omega).
\end{aligned}
$$

This integral has taken a sinusoid and compressed it into a delta function located at its frequency! If $f(t)$ is not a sinusoid, then this integral would essentially pick out the part of $f(t)$ that acts like a sinusoid at frequency ω. This is the Fourier transform! We often call the transformed function

$$
\boxed{\tilde{f}(\omega) \equiv \int_{-\infty}^{\infty} f(t) e^{-i\omega t} \, \mathrm{d}t}
$$

We can also invert the Fourier transform to get our orignal function back:

$$
\int_{-\infty}^{\infty} \tilde{f}(\omega) e^{+i\omega t} \, \mathrm{d}\omega = f(t)
$$

The Fourier transform is an **integral transform**. Basically, these relate one set of functions to another set of functions. There are literally *infinitely many* of these (more about this later).

The Fourier transform **is designed to measure the periodic behavior of functions!**

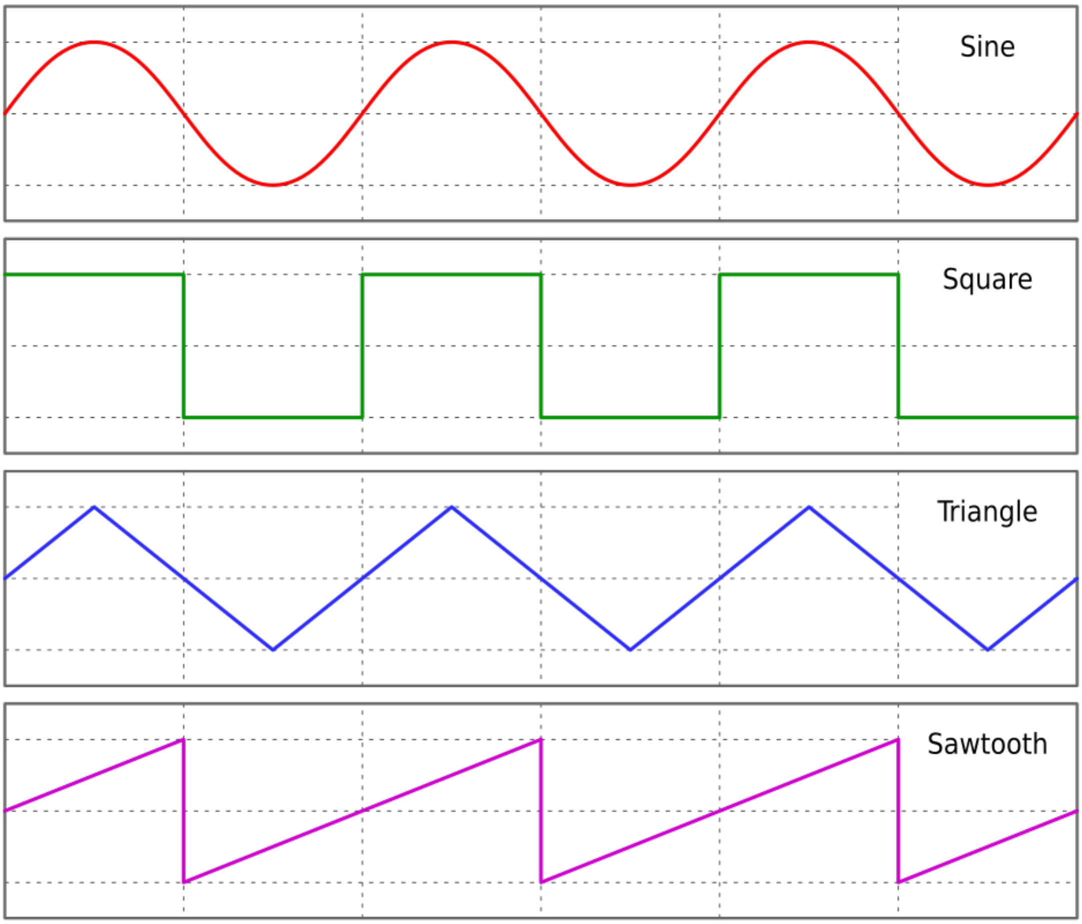
Great for answering questions like these:

- what pitch is that sound? (peak of fourier transform)
- I have noisy data, is there a sine wave in it? (peak of fourier transform)
- when will the tide rise? (model as sum of sinusoids -> moon, sun)
- Can't solve an equation? Solve it for a sine wave -> solution via Fourier

Some examples

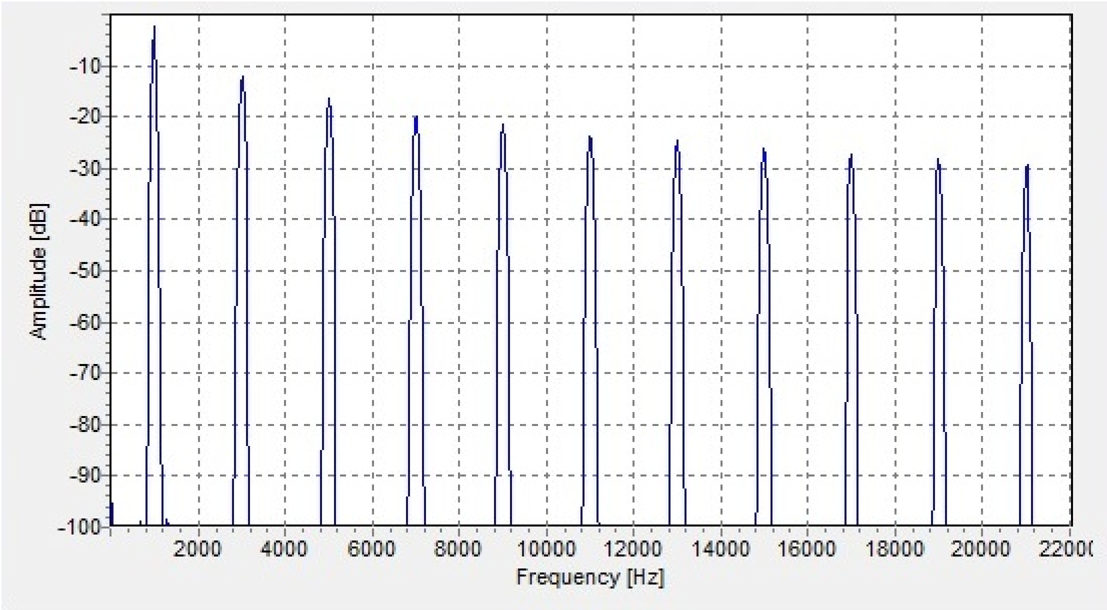
Common non-sine periodic things

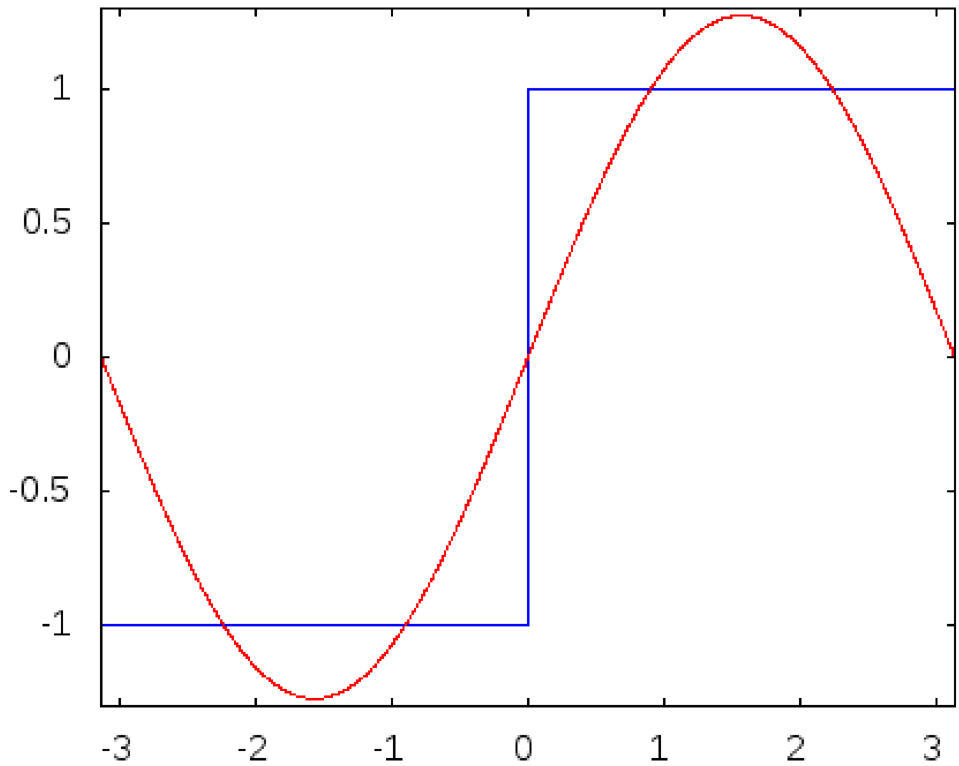
These signals are periodic, but not just a sine wave, it's a little more complicated. The Fourier spectrum then has a peak at the equivalent sine wave frequency but also more peaks at integer multiples of that fundamental frequency. These are called **harmonics**.



A square wave can be written:

$$\begin{aligned} x(t) &= \operatorname{sgn}\left(\sin \frac{2\pi t}{t_0}\right) \\ &= \frac{4}{\pi}\left(\sin(\omega t) + \frac{1}{3}\sin(3\omega t) + \frac{1}{5}\sin(5\omega t) + \ldots\right) \end{aligned}$$





Sharp corners: Gibbs phenomenon.

Square-ish wave

This signal is a square wave but only turned on about 10 percent of the time. It has a lot of harmonics

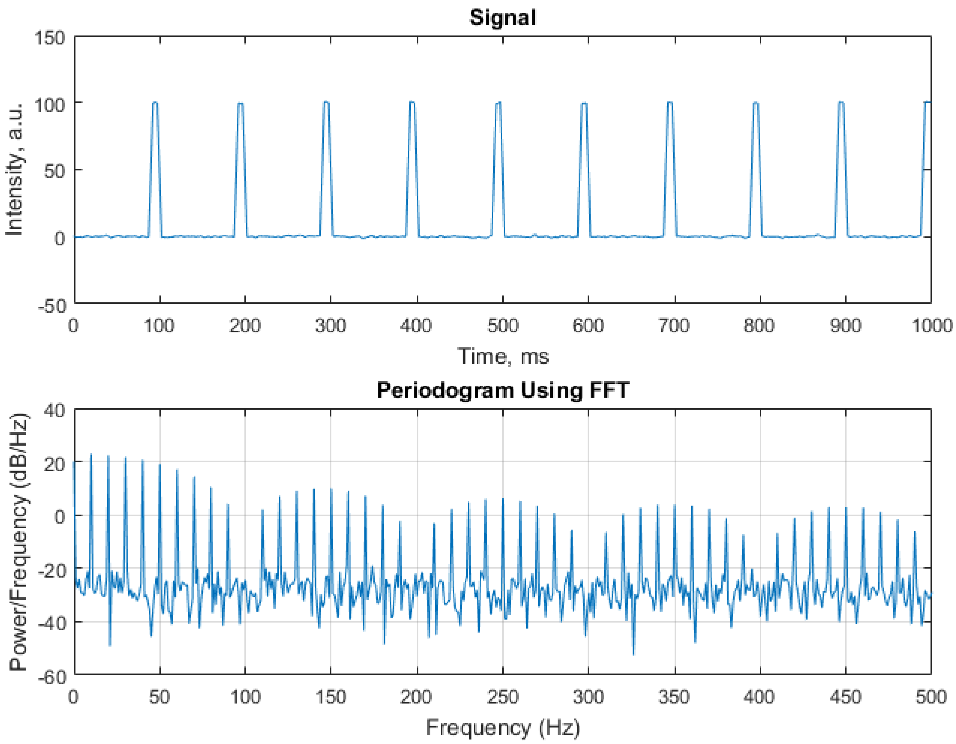
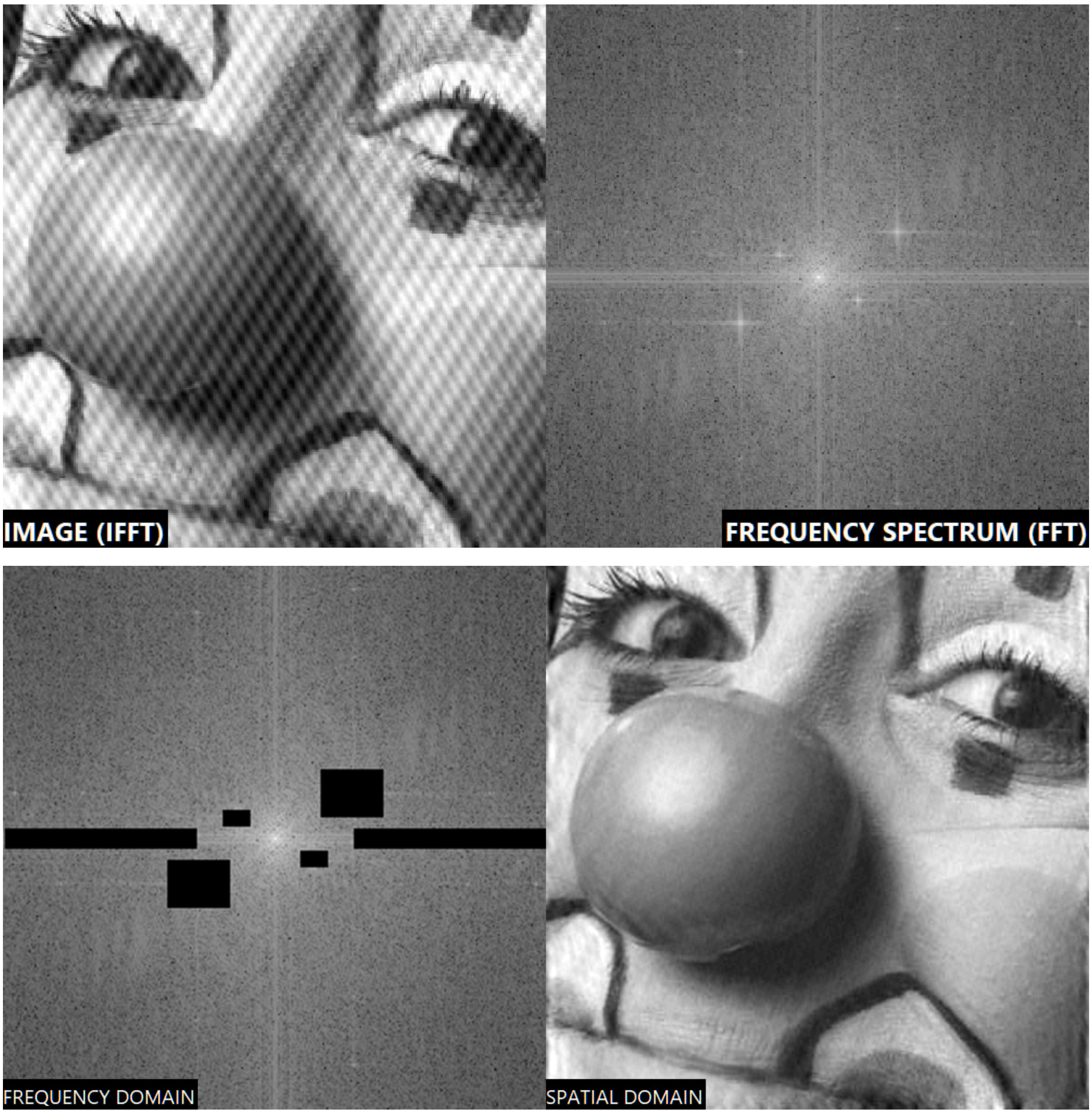


Figure 1: squareish_wave.png

Image Fourier Transform

The Fourier transform can also be applied to "spatial frequencies", or sine waves in distance, not time. These can be exploited to do a lot of cool image processing. It's also even possible to do this **physically** with Fourier optics. But let's check out a cool image processing technique with Fourier methods. If we have periodic noise in an image, this can show up as a bright point in its spatial Fourier transform. This can then be zeroed out and the transform inverted to get rid of the noise entirely!

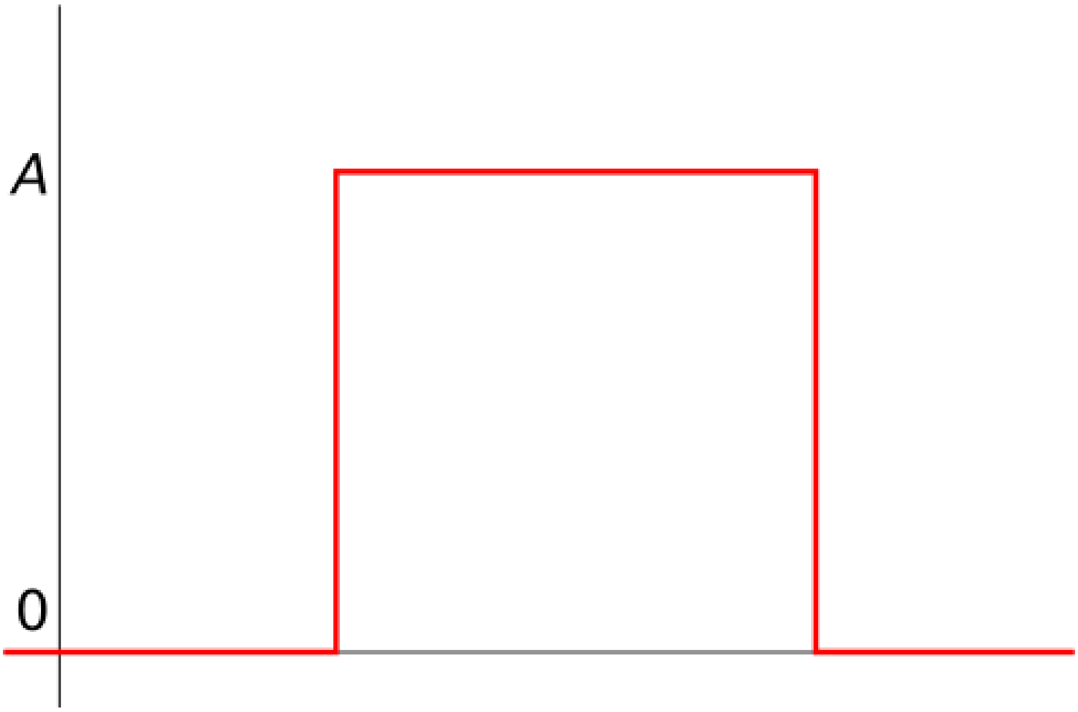
Check out https://blog.ampedsoftware.com/2023/10/03/remove-periodic-noise-from-an-image



Boxcar function (rectangle function)

This function represents signals turning on and off!

$$\Pi(t) \equiv \theta(t - 1/2) - \theta(t + 1/2)$$



It has a nice fourier transfom as a sinc function $\mathrm{sinc}(\omega/2) \equiv \sin(\omega/2)/(\omega/2)$. If we scale this to cover a window of total length T, then the Fourier transform of $\Pi(t/T)$ is just $T\mathrm{sinc}(T\omega/2)$.

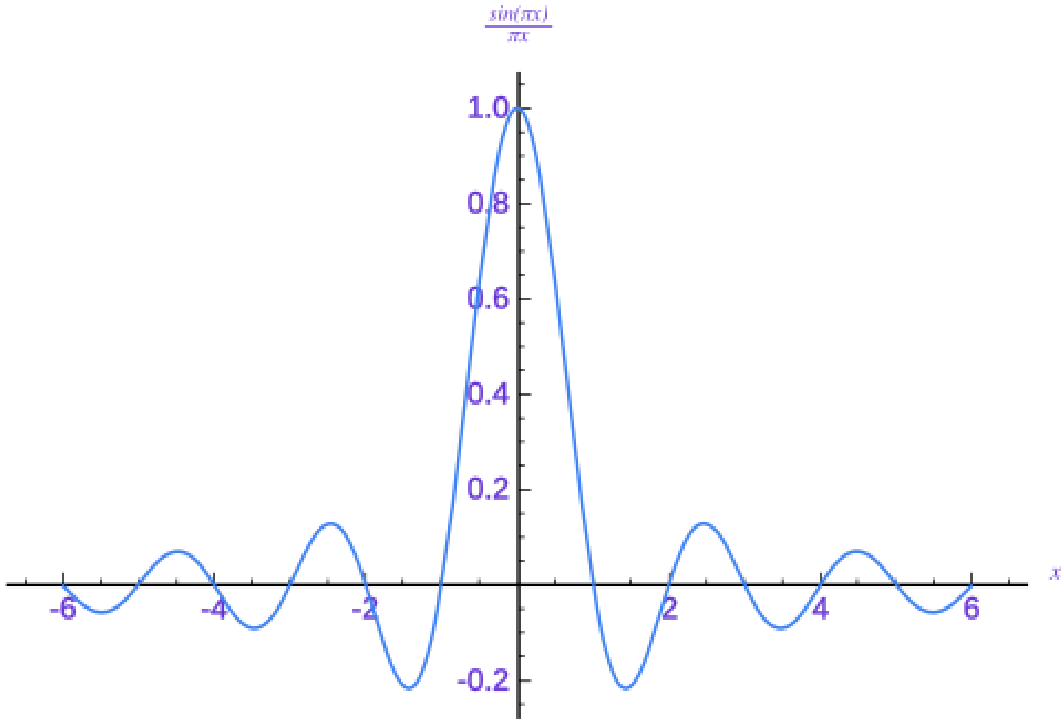
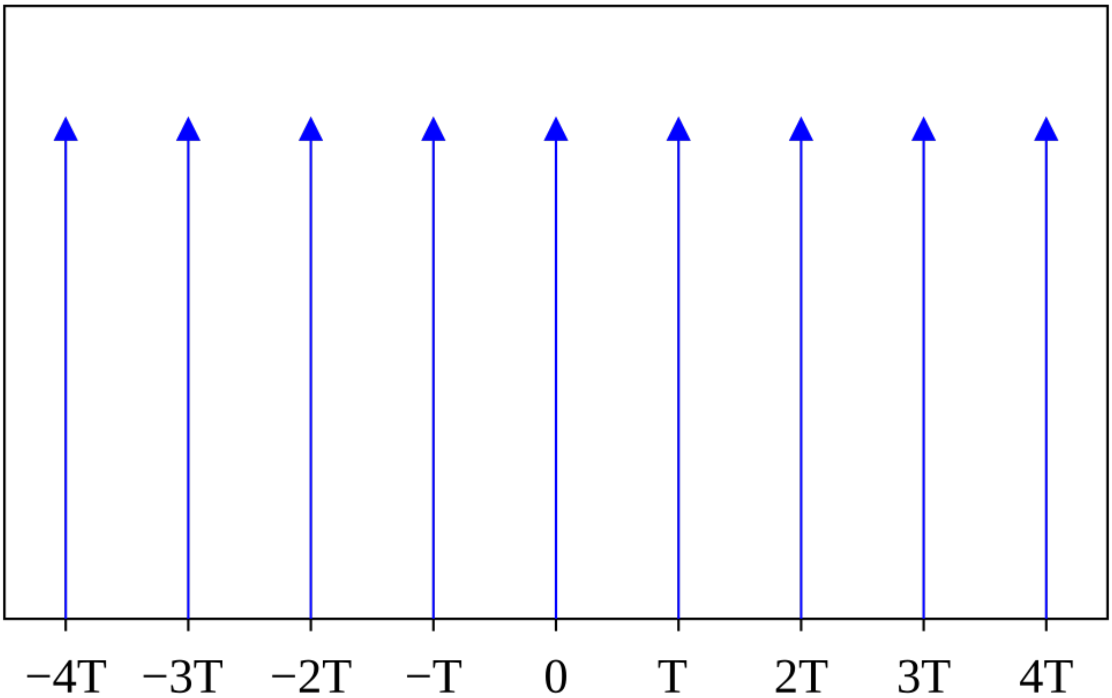


Figure 2: sinc.png

Comb function

This function is very useful for modeling sampled data. We can write it

$$\text{Ш}_T(t) \equiv \sum_{n=-\infty}^{\infty} \delta(t - nT)$$



The Fourier transform of a Dirac comb is another Dirac comb! Shrunken and scaled

$$\text{Ш}_T(t) \Rightarrow \frac{1}{T}\text{Ш}_{1/T}(\omega)$$

Gaussian

The fourier transform of a Gaussian is… a Gaussian!

$$x(t) = e^{-t^2/(2\sigma^2)} \to \tilde{x}(\omega) = \sqrt{2\pi\sigma^2}e^{-\omega^2\sigma^2/2}$$

Wide gaussian in time domain -> narrow in Fourier domain and vice versa.

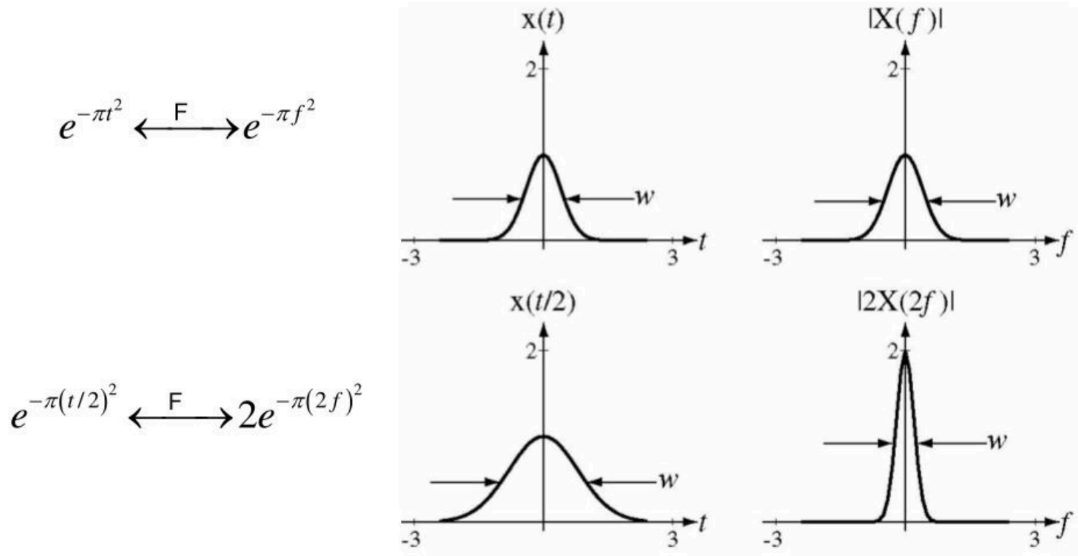


Figure 3: ./gaussian_uncertainty.png

Some important and cool facts about $\tilde{f}(\omega)$

Fourier uncertainty principle

The Gaussian above is a perfect example. Better time resolution? Worse frequency resolution (and vice-versa). Gaussians actually saturate this, but other signals can do worse.

$$\sigma_t \sigma_f \geq \frac{1}{4\pi}$$

This is actually the same uncertainty principle as in quantum mechanics!

Parseval's theorem

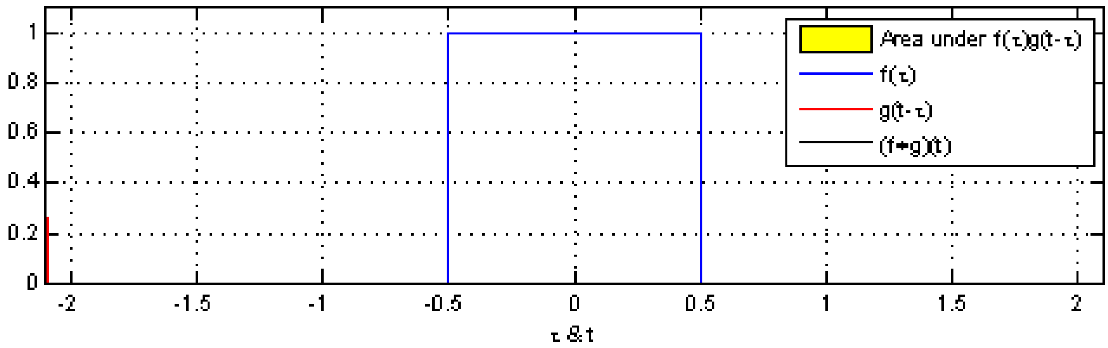
$$\int | \; \tilde{f}(\omega) \; |^2 \; d\omega = \int | \; f(t) \; |^2 \; dt$$

Total integral over the signal is the same in either domain. Signal energy is conserved across the Fourier transform.

Convolution theorem

This one is **really important**. Convolution in the time domain? Multiplication in Fourier domain. We can model many processes happing to data as **convolutions**:

$$f(t) \otimes g(t) \equiv \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau$$



If we try taking the Fourier transform of this and swap some integrals around:

$$\begin{aligned}
\int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} f(\tau)g(t-\tau)\mathrm{d}\tau\right]e^{-i\omega t}\mathrm{d}t &= \int_{-\infty}^{\infty} f(\tau)\left[\int_{-\infty}^{\infty} g(t-\tau)e^{-i\omega t}\mathrm{d}t\right]\mathrm{d}\tau \\
&= \int_{-\infty}^{\infty} f(\tau)\left[\int_{-\infty}^{\infty} g(u)e^{-i\omega(u+\tau)}\mathrm{d}u\right]\mathrm{d}\tau \\
&= \left[\int_{-\infty}^{\infty} f(\tau)e^{-i\omega\tau}\mathrm{d}\tau\right]\left[\int_{-\infty}^{\infty} g(u)e^{-i\omega u}\mathrm{d}u\right] \\
&= F(\omega)G(\omega)
\end{aligned}$$

We turned convolution in the time domain into multiplication in Fourier domain. This is very useful for thinking what Fourier transforms will do.

Deconvolution: we can try to do $F(\omega)/G(\omega)$ to undo a convolution. If we could do this perfectly, then we can undo any convolution operation. Real life examples include de-blurring an image.

Reciprocity theorem

This is more of a concept than a mathematical result. **Mathematically, there is very little difference between time and frequency domains.**

- Things like convolution theorem always work both ways
- If you know the transform one way, then you know it both ways
- The discreteness results we will get below go both ways. Limits in time domain give analagous limits in frequency domain.

Intuitive applications

- Gaussian train: this is a comb convolved with a gaussian. Then the FFT must be FT of comb * FT of gaussian.

$$x(t) = \text{Ш}_T(t) \otimes e^{-t^2/(2\sigma^2)}$$

The Fourier transform will be

$$\begin{aligned}\tilde{x}(\omega) &= \mathcal{F}\{\text{Ш}_T(t)\}\mathcal{F}\left\{e^{-t^2/(2\sigma^2)}\right\} \\ &= \frac{1}{T}\text{Ш}_{1/T}(\omega)\sqrt{2\pi\sigma^2}e^{-\omega^2\sigma^2/2} \\ &= \frac{\sqrt{2\pi\sigma^2}}{T}\text{Ш}_{1/T}(\omega)e^{-\omega^2\sigma^2/2}\end{aligned}$$

We get a smooth gaussian envelope, with peaks every $1/T$ in frequency.

- Impulse $\delta(t)$: equal power at all frequencies

$$\mathcal{F}(\delta(t)) = 1$$

$$\int_{-\infty}^{\infty} \delta(t)e^{-i\omega t}\,\mathrm{d}t = e^0 = 1$$

- Let's look again at our square-ish wave.

  Frequency of pulses is at $T = 1/(10\,\mathrm{Hz})$. Pulses are rectangle functions with width $0.01\,\mathrm{s}$.

$$\begin{aligned}x(t) &= \text{Ш}_T(t) \otimes \Pi\left(\frac{t}{0.01\,\mathrm{s}}\right) \\ \tilde{x}(\omega) &= \mathcal{F}\{\text{Ш}_T(t)\} \cdot (0.01\,\mathrm{s})\mathrm{sinc}((0.01\,\mathrm{s})\omega/2) \\ &= \frac{1}{T}\text{Ш}_{1/T}(\omega) \cdot (0.01\,\mathrm{s})\mathrm{sinc}((0.01\,\mathrm{s})\omega/2)\end{aligned}$$

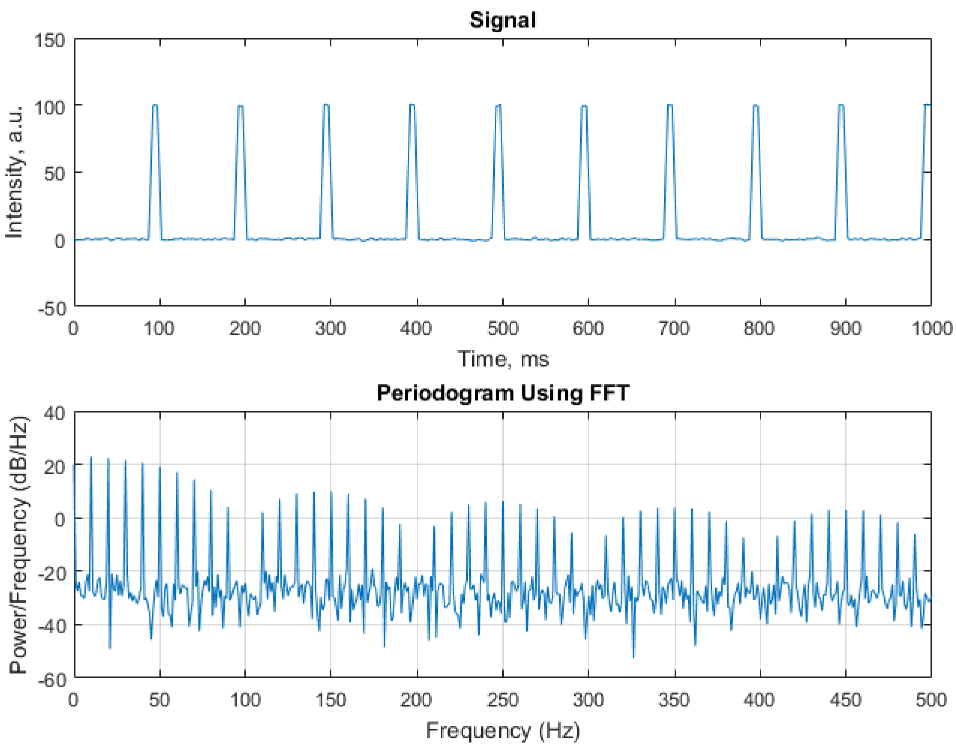


Figure 4: squareish_wave.png

See this page for more details on this intuitive explanation of this signal: https://dsp.stackexchange.com/questions/58888/harmonics-of-square-wave